Computer Networks Lab 07

Course: Computer Networks (CL3001) | Semester: Fall 2026
Course Coordinator: Ms. Ramsha Iqbal

Learning Objectives

- To understand the concept, purpose, and working mechanism of Access Control Lists (ACLs) in network security.
- To differentiate between Standard and Extended ACLs and understand their placement rules (inbound/outbound and source/destination proximity).
- To learn to configure, apply and verify ACLs on Cisco routers to filter network traffic.

ACL (Standard + Extended (Basic))

1. Introduction to Access Control List (ACL)

Access-list (ACL) is a set of rules defined for controlling network traffic and reducing network attacks. ACLs are used to filter traffic based on the set of rules defined for the incoming or outgoing of the network. ACLs are mainly found in network devices with packet filtering capabilities including routers and switches.

Different ACLs have different working mechanisms based on what they do. For File system ACLs, they work by creating tables that inform the operating system of access privileges given for certain system subjects. Each object has a unique security property that acts as an identification factor in its access control list. Some privileges include read/write privileges, file execution, and several others.

Some popular operating systems utilizing this mechanism include Unix-based systems, Windows NT/2000, and Novell's Netware.

In the case of Networking ACLS, they are installed in networking devices (Routers and switches) with the sole purpose of filtering traffic. This is done by using pre-defined rules that decided which packets transferred. Source and destination IP addresses also play a major role in this decision.

Packet filtering improves network security by decreasing network traffic access, restricting device and user access to the involved network.

Access lists are sequential, and are made up of two major components; permit and deny statements. A name and a number are used to identify access lists.

Features of ACL

- The set of rules defined are matched serial wise i.e. matching starts with the first line, then 2nd, then 3rd, and so on.
- The packets are matched only until it matches the rule. Once a rule is matched then no further comparison takes place and that rule will be performed.
- There is an implicit denial at the end of every ACL, i.e., if no condition or rule matches then the packet will be discarded.

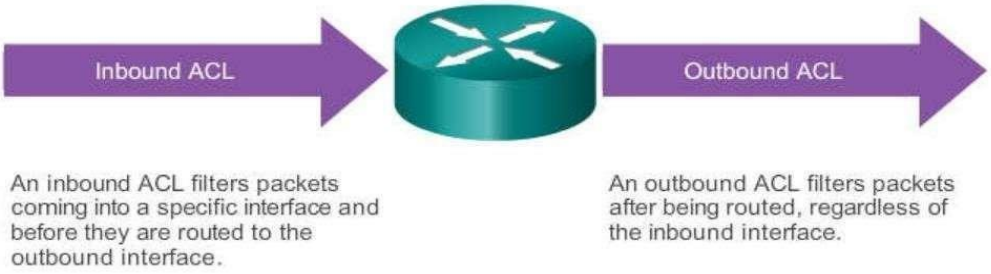
Once the access-list is built, then it should be applied to inbound or outbound of the interface.

Inbound access lists

When an access list is applied on inbound packets of the interface then first the packets will be processed according to the access list and then routed to the outbound interface.

Outbound access lists

When an access list is applied on outbound packets of the interface then first the packet will be routed and then processed at the outbound interface.



Types of ACL

- **Standard ACL:** A Standard ACL filters traffic based on the source IP address only. It permits or denies the entire protocol suite and cannot differentiate between traffic types such as TCP, UDP, or ICMP. It uses the number range 1-99 and 1300-1999.
- **Extended ACL:** An Extended ACL filters traffic based on source IP, destination IP, source port, destination port, and protocol type (e.g., TCP, UDP, ICMP). This allows for more granular control over specific traffic. It uses the number range 100-199 and 2000-2699.
- **Reflexive ACL:** Also known as IP session ACLs, Reflective ACLs use upper-layer session details to filter traffic.
- **Dynamic ACL:** As the term suggests, Dynamic ACLs are reliable on extended ACLs, Telnet, and authentication. They grant users access to a resource only if the user authenticates the device through telnet.

Also, there are two categories of access-list:

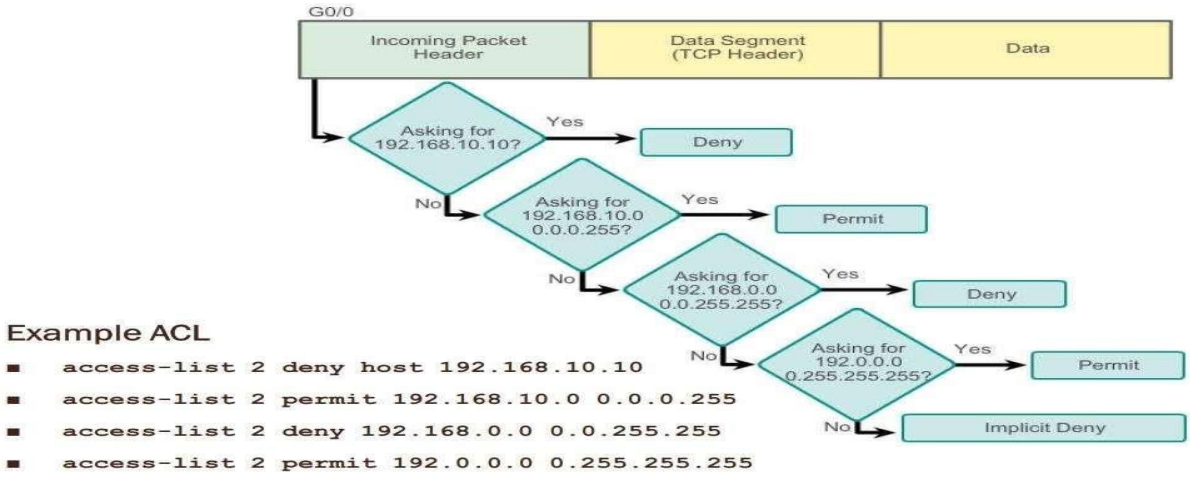
- **Numbered Access-List:** These are the access list that cannot be deleted specifically once created i.e., if we want to remove any rule from an Access-list then this is not permitted in the case of the numbered access list. If we try to delete a rule from the access list, then the whole access list will be deleted. The numbered access-list can be used with both standard and extended access lists.
- **Named Access-List:** In this type of access list, a name is assigned to identify an access list. It is allowed to delete a named access list, unlike numbered access list. Like numbered access lists, these can be used with both standards and extended access lists.

Advantages of ACL

- Improve network performance.
- Provides security as the administrator can configure the access list according to the needs and deny the unwanted packets from entering the network.
- Provides control over the traffic as it can permit or deny according to the need of the network.

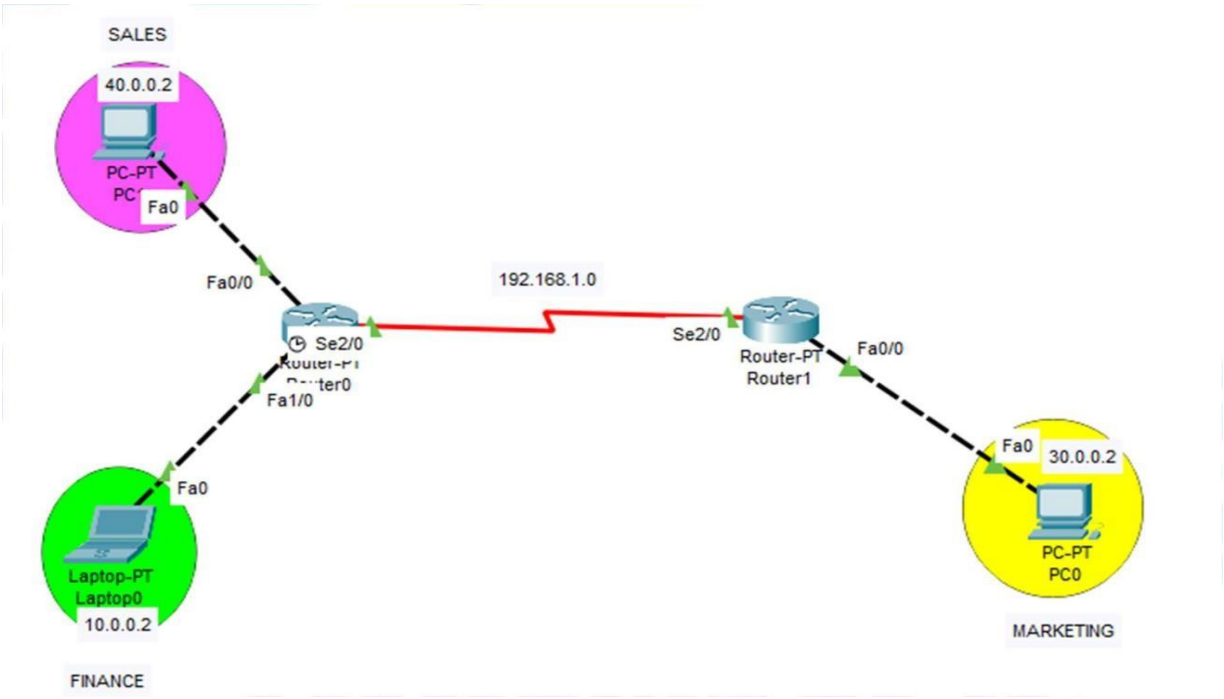
Rules of ACL

- The standard Access-list is generally applied close to the destination (but not always).
- The extended Access-list is generally applied close to the source (but not always).
- We can assign only one ACL per interface per protocol per direction, i.e., only one inbound and outbound ACL is permitted per interface.
- We can't remove a rule from an Access-list if we are using numbered Access-list. If we try to remove a rule, then the whole ACL will be removed. If we are using named access lists, then we can delete a specific rule.
- Every new rule which is added to the access list will be placed at the bottom of the access list therefore before implementing the access lists, analyses the whole scenario carefully.
- As there is an implicit deny at the end of every access list, we should have at least a permit statement in our Access-list otherwise all traffic will be denied.
- Standard access lists and extended access lists cannot have the same name.

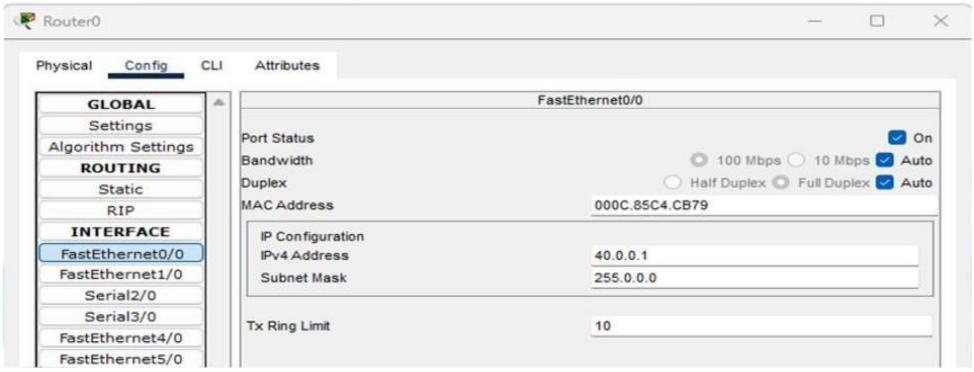


2. Implementation of Extended Access Control List (ACL)

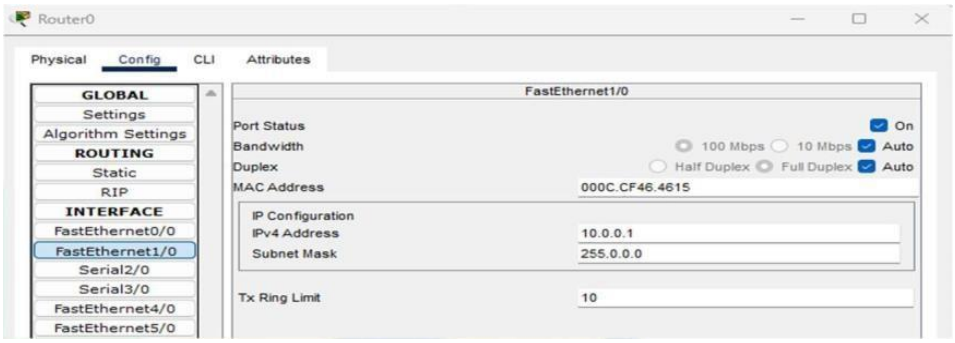
Consider the topology given below:



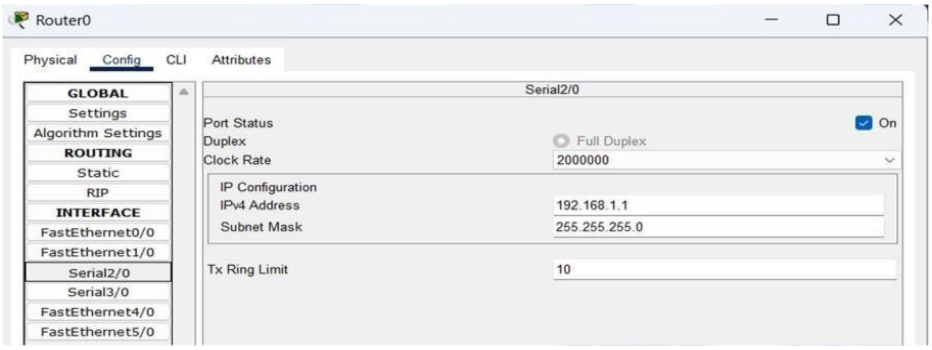
IP configuration on Router 0 (fa0/0):



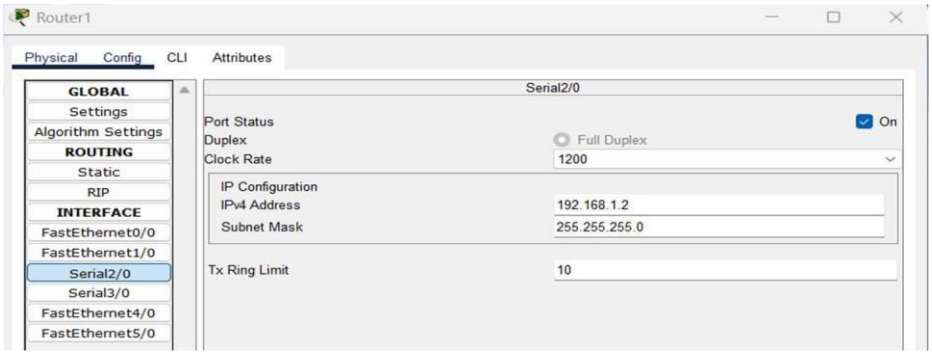
IP configuration on Router 0 (fa0/1):



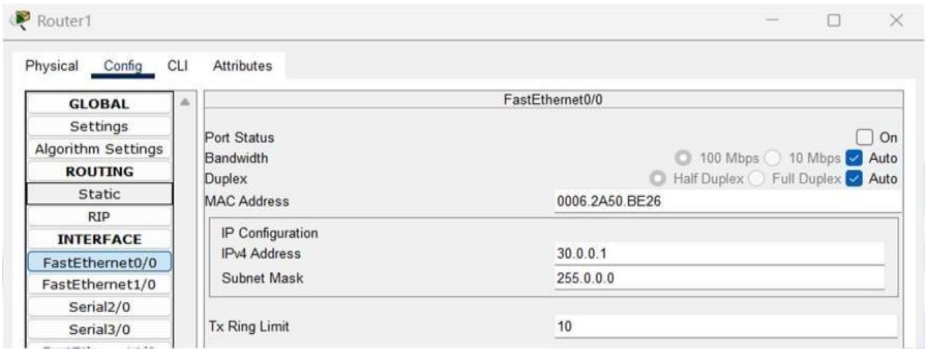
IP configuration on Router 0 (se2/0):



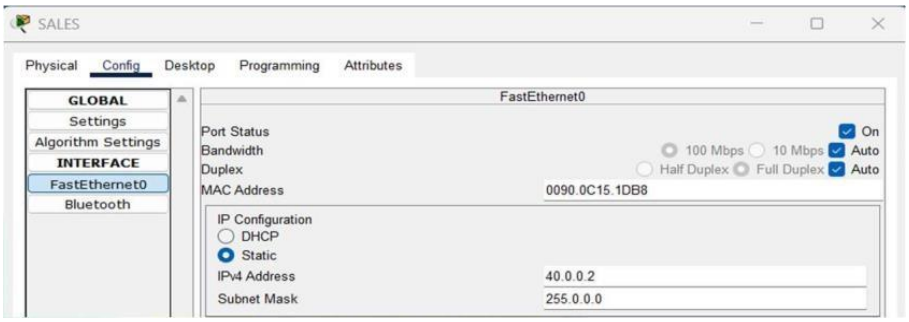
IP configuration on Router 1 (se2/0):



IP configuration on Router 1 (fa0/0):



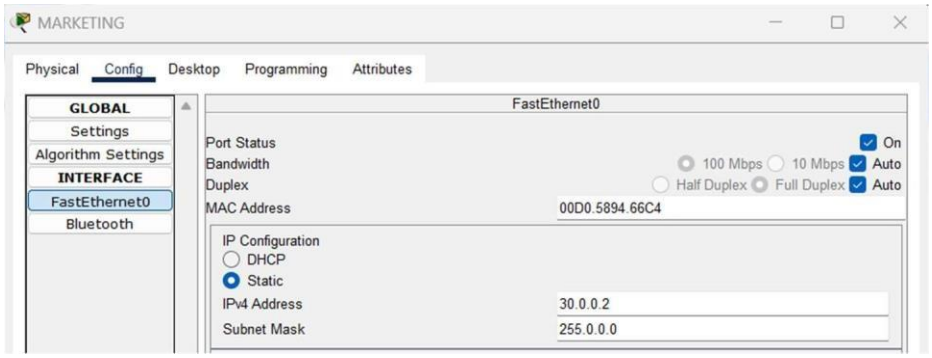
IP configuration on PC (SALES):



IP configuration on PC (FINANCE):



IP configuration on PC (MARKETING):



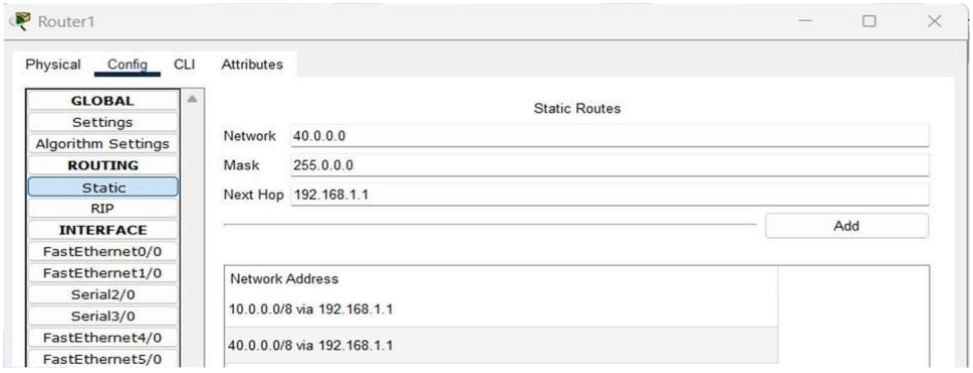
Configure static routing on both routers (specify serial port IPs in static routing):

Router 0:



Router 1:



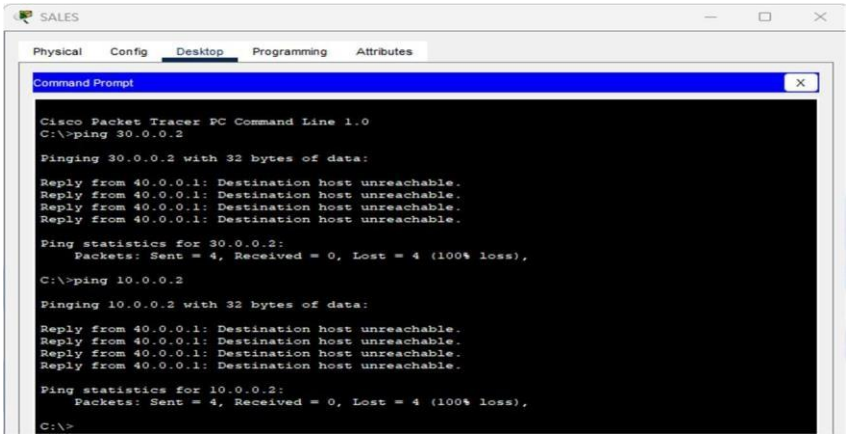


❖ Block All ICMP (PING & Other Messages):

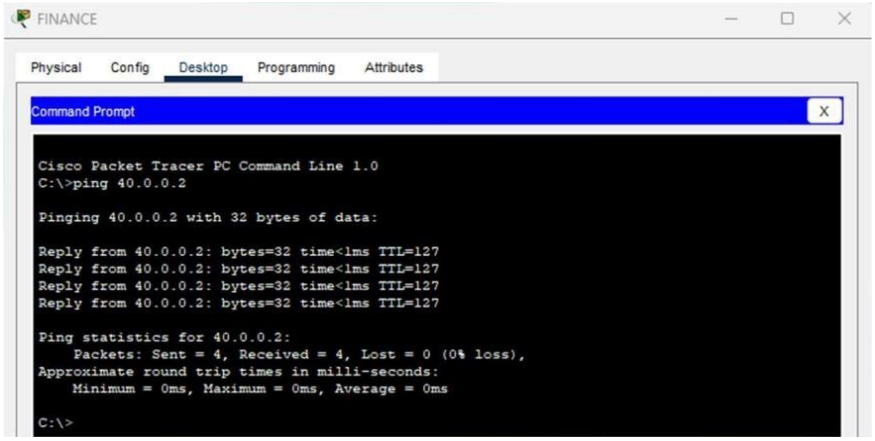
Suppose we want to block SALES to ping any other PCs (MARKETING & FINANCE). But all other PCs must be allowed to ping SALES:

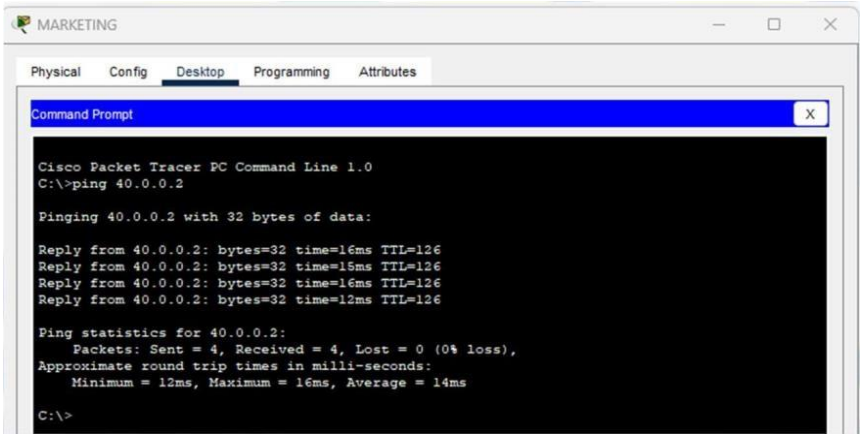
```
Router(config)#ip access-list extended PING-ONEWAY
Router(config-ext-nacl)#remark Block ping from PC1 (40.0.0.2) to PC2 (10.0.0.2)
Router(config-ext-nacl)#deny icmp host 40.0.0.2 host 10.0.0.2 echo
Router(config-ext-nacl)#deny icmp host 40.0.0.2 host 30.0.0.2 echo
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#
Router(config-ext-nacl)#exit
```

```
Router(config)#interface FastEthernet 0/0
Router(config-if)#ip access-group PING-ONEWAY in
Router(config-if)#
Router(config-if)#exit
```



As you can see, SALES is now unable to ping FINANCE & MARKETING. Now let`s try pinging SALES from other hosts:

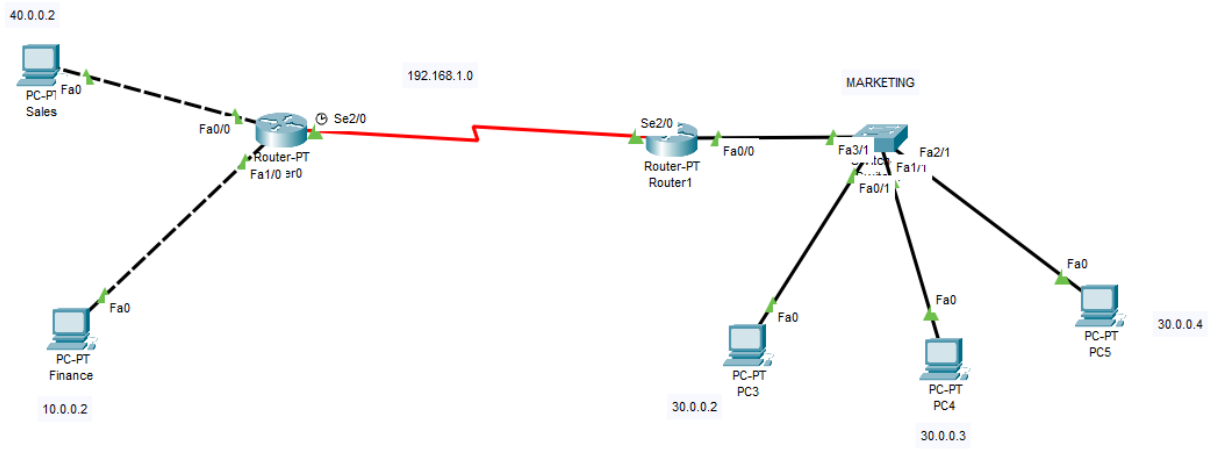




All other hosts are now able to ping SALES.

❖ Block a Whole Network:

Consider the topology given below:



Wildcard Mask:

Wildcard mask is used in Access Control Lists (ACL) and routing protocols like OSPF to define a range of IP addresses. It is the inverse of a subnet mask. While a subnet mask uses 255 to indicate the network portion, a wildcard mask uses 0 to indicate that the corresponding octet must match exactly, and 255 to indicate that the octet can be ignored. For example, the subnet mask 255.0.0.0 for a /8 network has a wildcard mask of 0.255.255.255.

Wildcard = 255.255.255.255 - Subnet Mask

Wildcard = 255.255.255.255 - 255.0.0.0

Wildcard = 0.255.255.255

Suppose we want to block the entire network of MARKETING (30.0.0.0/8) to be accessed from FINANCE:

On router1:

```
Router(config)#ip access-list extended MRK-BLOCK
Router(config-ext-nacl)#remark block entire FINANCE 10.0.0.0/8 to MARKETING 30.0.0.0/8
Router(config-ext-nacl)#deny icmp 10.0.0.0 0.255.255.255 30.0.0.0 0.255.255.255 echo
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#ip access-group MRK-BLOCK out
Router(config-if)#exit
```

Initially, the Extended ACL was applied on Router1 (destination network) in the outbound direction. In some cases, Packet Tracer may not filter traffic as expected when the ACL is applied outbound at the destination. To follow best practice and ensure proper filtering, it was moved closer to the source on Router0 interface Fa1/0 inbound, so traffic from FINANCE to MARKETING is dropped at the source.

On router0:

```
Router(config)#ip access-list extended MRK-BLOCK
Router(config-ext-nacl)#remark block entire FINANCE 10.0.0.0/8 to MARKETING 30.0.0.0/8
Router(config-ext-nacl)#deny icmp 10.0.0.0 0.255.255.255 30.0.0.0 0.255.255.255 echo
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit
```

```
Router(config)#interface FastEthernet 1/0
Router(config-if)#ip access-group MRK-BLOCK in
Router(config-if)#exit
```

Ping from MARKETING to FINANCE Network (Successful):

```
C:\>ping 10.0.0.0

Pinging 10.0.0.0 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=3ms TTL=254
Reply from 192.168.1.1: bytes=32 time=1ms TTL=254
Reply from 192.168.1.1: bytes=32 time=2ms TTL=254
Reply from 192.168.1.1: bytes=32 time=1ms TTL=254

Ping statistics for 10.0.0.0:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\>
```

Ping from FINANCE to MARKETING Network (Failed):

```
C:\>ping 30.0.0.0

Pinging 30.0.0.0 with 32 bytes of data:

Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.

Ping statistics for 30.0.0.0:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```